

## CHISHOLM-HIBBING, MN, USA

WMO: 727455

Lat: 47.386N

Lon: 92.839W

Elev: 1352

StdP: 13.99

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94931

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -24.0      | -18.2 | -29.2                         | 1.1 | -23.2 | -24.2 | 1.5 | -17.3 | 24.7                  | 8.6  | 22.5 | 8.6  | 3.4                      | 320  | 0.634 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 24.8                   | 85.2            | 69.7 | 82.2 | 67.7 | 79.8 | 65.9 | 72.9                | 82.1 | 70.4 | 78.9 | 68.3 | 76.2 | 9.7                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.8                            | 115.1 | 78.4 | 67.4 | 106.1 | 75.7 | 65.3 | 98.4 | 73.1 | 37.4          | 82.2 | 35.2 | 78.6 | 33.3 | 76.1 | 80.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |  |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|--|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |  |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |  |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |  |
| 20.9              | 18.7 | 16.8 | DB                         | -31.6 | 89.9               | 6.4 | 2.8                                                | -36.3 | 92.0       | -40.0 | 93.6       | -43.6 | 95.2       | -48.3 | 97.2 |  |
|                   |      |      | WB                         | -30.9 | 76.5               | 5.6 | 1.7                                                | -34.9 | 77.7       | -38.2 | 78.8       | -41.3 | 79.7       | -45.3 | 81.0 |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 38.6   | 7.6   | 11.9  | 25.0  | 38.6  | 50.8  | 60.5  | 65.2  | 63.0  | 55.0  | 42.3  | 27.5  | 13.8  |
|                                                                                  | DBStd       | 22.40  | 14.13 | 13.30 | 12.74 | 9.17  | 8.13  | 6.69  | 5.71  | 5.92  | 8.26  | 8.29  | 10.75 | 12.84 |
|                                                                                  | HDD50       | 5715   | 1315  | 1067  | 778   | 354   | 87    | 4     | 0     | 0     | 41    | 269   | 677   | 1123  |
|                                                                                  | HDD65       | 9822   | 1780  | 1487  | 1241  | 791   | 446   | 167   | 69    | 109   | 313   | 705   | 1126  | 1588  |
|                                                                                  | CDD50       | 1541   | 0     | 0     | 2     | 13    | 112   | 318   | 472   | 403   | 192   | 28    | 1     | 0     |
|                                                                                  | CDD65       | 173    | 0     | 0     | 0     | 0     | 6     | 31    | 75    | 47    | 14    | 0     | 0     | 0     |
|                                                                                  | CDH74       | 2411   | 0     | 0     | 0     | 7     | 167   | 470   | 969   | 603   | 182   | 13    | 0     | 0     |
|                                                                                  | CDH80       | 533    | 0     | 0     | 0     | 0     | 33    | 107   | 235   | 125   | 32    | 1     | 0     | 0     |
| Wind                                                                             | WSAvg       | 7.1    | 7.4   | 7.7   | 7.8   | 8.2   | 7.7   | 6.6   | 6.0   | 5.7   | 6.3   | 7.3   | 7.4   | 7.1   |
| Precipitation                                                                    | PrecAvg     | 25.4   | 0.7   | 0.5   | 0.9   | 1.6   | 2.8   | 3.9   | 4.0   | 3.4   | 3.0   | 2.3   | 1.2   | 0.7   |
|                                                                                  | PrecMax     | 32.6   | 2.3   | 1.6   | 2.3   | 4.4   | 5.6   | 6.6   | 9.6   | 7.2   | 6.4   | 5.7   | 3.6   | 1.5   |
|                                                                                  | PrecMin     | 17.5   | 0.0   | 0.0   | 0.1   | 0.2   | 0.4   | 0.8   | 1.0   | 0.6   | 0.5   | 0.4   | 0.1   | 0.0   |
|                                                                                  | PrecStd     | 4.0    | 0.5   | 0.4   | 0.6   | 1.0   | 1.3   | 1.6   | 2.0   | 1.8   | 1.5   | 1.5   | 0.9   | 0.4   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 40.4  | 45.3  | 63.4  | 73.3  | 84.3  | 88.2  | 89.3  | 88.2  | 83.9  | 75.4  | 59.7  | 42.0  |
|                                                                                  |             | MCWB   | 35.0  | 39.0  | 51.0  | 55.6  | 64.7  | 72.7  | 73.4  | 72.6  | 68.8  | 59.8  | 50.8  | 36.5  |
|                                                                                  | 2%          | DB     | 34.3  | 39.6  | 53.8  | 66.9  | 79.0  | 82.6  | 85.6  | 83.2  | 79.3  | 67.8  | 53.0  | 37.0  |
|                                                                                  |             | MCWB   | 31.8  | 35.6  | 44.6  | 50.5  | 60.3  | 67.3  | 70.2  | 69.1  | 66.2  | 55.9  | 46.5  | 34.2  |
|                                                                                  | 5%          | DB     | 31.2  | 35.9  | 47.7  | 62.0  | 73.5  | 79.4  | 82.3  | 80.5  | 75.0  | 62.6  | 47.5  | 34.1  |
|                                                                                  |             | MCWB   | 29.6  | 32.7  | 40.3  | 47.4  | 57.7  | 65.1  | 68.6  | 67.0  | 63.5  | 54.3  | 41.8  | 32.1  |
|                                                                                  | 10%         | DB     | 27.7  | 31.9  | 42.5  | 56.6  | 68.7  | 75.8  | 79.7  | 77.2  | 71.0  | 57.0  | 43.0  | 31.6  |
|                                                                                  |             | MCWB   | 25.9  | 29.4  | 37.1  | 44.8  | 54.9  | 62.8  | 66.7  | 65.0  | 61.8  | 50.2  | 38.9  | 30.2  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 35.4  | 39.9  | 54.6  | 58.0  | 68.2  | 74.5  | 75.9  | 74.8  | 71.4  | 63.1  | 53.3  | 37.9  |
|                                                                                  |             | MCDB   | 39.0  | 43.8  | 59.9  | 69.3  | 78.8  | 84.3  | 85.3  | 84.1  | 80.3  | 70.4  | 56.7  | 39.8  |
|                                                                                  | 2%          | WB     | 32.3  | 36.5  | 45.8  | 52.2  | 63.4  | 70.4  | 73.4  | 71.7  | 68.4  | 58.8  | 46.9  | 34.5  |
|                                                                                  |             | MCDB   | 34.3  | 39.6  | 51.3  | 64.2  | 74.4  | 80.2  | 82.9  | 80.2  | 75.7  | 66.0  | 51.9  | 36.7  |
|                                                                                  | 5%          | WB     | 29.4  | 32.7  | 41.3  | 48.6  | 60.0  | 67.5  | 70.9  | 69.1  | 65.9  | 54.9  | 42.5  | 32.3  |
|                                                                                  |             | MCDB   | 31.1  | 35.5  | 47.2  | 60.0  | 70.0  | 75.3  | 79.1  | 76.9  | 72.7  | 61.2  | 46.9  | 33.9  |
|                                                                                  | 10%         | WB     | 26.1  | 29.2  | 37.2  | 45.4  | 56.9  | 64.8  | 68.5  | 67.1  | 63.3  | 50.9  | 38.8  | 29.9  |
|                                                                                  |             | MCDB   | 27.8  | 31.8  | 42.2  | 55.4  | 66.4  | 72.7  | 76.5  | 74.6  | 69.3  | 56.4  | 42.5  | 31.5  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 19.8  | 21.9  | 21.6  | 23.1  | 25.9  | 24.6  | 24.8  | 24.7  | 23.8  | 19.5  | 16.2  | 16.7  |
|                                                                                  |             | MCDBR  | 18.8  | 23.0  | 26.7  | 35.0  | 35.0  | 29.5  | 28.3  | 28.4  | 29.7  | 30.3  | 24.6  | 18.7  |
|                                                                                  |             | MCWBR  | 16.9  | 18.4  | 18.1  | 20.8  | 19.1  | 16.3  | 15.1  | 15.5  | 17.7  | 19.3  | 17.6  | 16.0  |
|                                                                                  | 5% WB       | MCDBR  | 18.6  | 20.7  | 25.1  | 31.3  | 29.1  | 24.4  | 24.9  | 24.5  | 23.8  | 26.4  | 22.4  | 17.2  |
|                                                                                  | MCWBR       | 16.7   | 16.9  | 17.5  | 19.7  | 17.2  | 15.1  | 14.8  | 14.9  | 15.2  | 18.4  | 17.0  | 15.1  |       |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.247  | 0.260 | 0.308 | 0.347 | 0.364 | 0.371 | 0.387 | 0.376 | 0.352 | 0.308 | 0.283 | 0.261 |       |
|                                                                                  | taud        | 2.372  | 2.340 | 2.358 | 2.372 | 2.383 | 2.385 | 2.357 | 2.417 | 2.458 | 2.572 | 2.521 | 2.401 |       |
|                                                                                  | Ebn at Noon | 272    | 292   | 291   | 288   | 285   | 283   | 276   | 275   | 272   | 272   | 257   | 251   |       |
|                                                                                  | Edh at Noon | 22     | 28    | 33    | 36    | 37    | 37    | 37    | 34    | 30    | 23    | 19    | 19    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 455    | 814   | 1180  | 1473  | 1667  | 1782  | 1915  | 1618  | 1204  | 720   | 418   | 321   |       |
|                                                                                  | RadStd      | 52     | 76    | 111   | 148   | 121   | 134   | 85    | 123   | 93    | 98    | 53    | 33    |       |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (38 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page